

## ► PHARMACOLOGY—TOXICITIES AND ADVERSE EFFECTS

**Ingested seafood toxins** Toxin actions include **h**istamine release, **t**otal block of Na<sup>+</sup> channels, or opening of Na<sup>+</sup> channels to **c**ause depolarization.

| TOXIN                                  | SOURCE                                                               | ACTION                                                                                                       | SYMPTOMS                                                                                                                                         | TREATMENT                                   |
|----------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Histamine (scombroid poisoning)</b> | Spoiled dark-meat fish such as tuna, mahi-mahi, mackerel, and bonito | Bacterial histidine decarboxylase converts histidine to histamine<br>Frequently misdiagnosed as fish allergy | Mimics anaphylaxis: oral burning sensation, facial flushing, erythema, urticaria, itching; may progress to bronchospasm, angioedema, hypotension | Antihistamines<br>Albuterol +/- epinephrine |
| <b>Tetrodotoxin</b>                    | Pufferfish                                                           | Binds fast voltage-gated Na <sup>+</sup> channels in nerve tissue, preventing depolarization                 | Nausea, diarrhea, paresthesias, weakness, dizziness, loss of reflexes                                                                            | Supportive                                  |
| <b>Ciguatoxin</b>                      | Reef fish such as barracuda, snapper, and moray eel                  | Opens Na <sup>+</sup> channels, causing depolarization                                                       | Nausea, vomiting, diarrhea; perioral numbness; reversal of hot and cold sensations; bradycardia, heart block, hypotension                        | Supportive                                  |

**Age-related changes in pharmacokinetics** Aging alters the passage of drugs through the body and standard doses can result in ↑ plasma concentrations. Older patients often require reduced doses to prevent toxicity.

- Absorption—mostly unaffected.
- Distribution—↓ total body water (↓ V<sub>d</sub> of hydrophilic drugs → ↑ concentration), ↑ total body fat (↑ V<sub>d</sub> of lipophilic drugs → ↑ half-life).
- Metabolism—↓ hepatic mass and blood flow → ↓ first-pass metabolism, ↓ hepatic clearance. Phase I of drug metabolism is decreased; phase II is relatively preserved.
- Excretion—↓ renal mass and blood flow (↓ GFR) → ↓ renal clearance.

**Specific toxicity treatments**

| TOXIN                                             | TREATMENT                                                                                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Acetaminophen                                     | N-acetylcysteine (replenishes glutathione)                                                                                                    |
| AChE inhibitors, organophosphates                 | Atropine > pralidoxime                                                                                                                        |
| Antimuscarinic, anticholinergic agents            | Physostigmine (crosses BBB), control hyperthermia                                                                                             |
| Arsenic                                           | Dimercaprol, succimer                                                                                                                         |
| Benzodiazepines                                   | Flumazenil                                                                                                                                    |
| β-blockers                                        | Atropine, glucagon, saline                                                                                                                    |
| Carbon monoxide                                   | 100% O <sub>2</sub> , hyperbaric O <sub>2</sub>                                                                                               |
| <b>Copper</b>                                     | “ <b>Penny</b> ”cillamine (penicillamine), <b>trientine</b> ( <b>3 copper pennies</b> )                                                       |
| Cyanide                                           | Hydroxocobalamin, nitrites + sodium thiosulfate                                                                                               |
| Dabigatran                                        | Idarucizumab                                                                                                                                  |
| Digoxin                                           | Digoxin-specific antibody fragments                                                                                                           |
| Direct factor <b>Xa</b> inhibitors (eg, apixaban) | Ande <b>x</b> anet alfa                                                                                                                       |
| Heparin                                           | Protamine sulfate                                                                                                                             |
| Iron ( <b>Fe</b> )                                | De <b>fer</b> oxamine, de <b>fer</b> asirox, de <b>fer</b> iprone                                                                             |
| Lead                                              | <b>P</b> enicillamine, calcium disodium <b>E</b> DTA, <b>D</b> imercaprol, <b>S</b> uccimer, (correct lead poisoning in <b>PEDS</b> patients) |
| <b>Mercury</b>                                    | <b>D</b> imercaprol, <b>succimer</b>                                                                                                          |
| Methanol, ethylene glycol (antifreeze)            | Fomepizole > ethanol, dialysis                                                                                                                |
| <b>Methemoglobin</b>                              | <b>M</b> ethylene blue, vitamin C (reducing agent)                                                                                            |
| Methotrexate                                      | Leucovorin                                                                                                                                    |
| <b>Opioids</b>                                    | <b>Naloxone</b>                                                                                                                               |
| Salicylates                                       | NaHCO <sub>3</sub> (alkalinizes urine), dialysis                                                                                              |
| TCAs                                              | NaHCO <sub>3</sub> (stabilizes cardiac cell membrane)                                                                                         |
| Warfarin                                          | Vitamin K (delayed effect), PCC (prothrombin complex concentrate)/FFP (immediate effect)                                                      |

**Drug reactions—cardiovascular**

| DRUG REACTION                 | CAUSAL AGENTS                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coronary vasospasm            | <b>C</b> ocaine, <b>A</b> mphetamines, <b>S</b> umatriptan, <b>E</b> rgot alkaloids ( <b>CASE</b> )                                                                                                                                                                                                                                                                                             |
| Cutaneous <b>flushing</b>     | <b>V</b> ancomycin, <b>A</b> denosine, <b>N</b> iacin, <b>Ca</b> <sup>2+</sup> channel blockers, <b>E</b> chinocandins, <b>N</b> itrates ( <b>flushed</b> from <b>VANCEN</b> [dancing])<br><b>Vancomycin infusion reaction</b> (formerly called red man syndrome)—histamine release → widespread pruritic erythema. Infusion rate–dependent. Manage with slower infusion rate, diphenhydramine. |
| <b>Dilated cardiomyopathy</b> | Alcohol, anthracycline (eg, <b>d</b> oxorubicin, <b>d</b> aunorubicin; prevent with <b>d</b> exrazoxane), trastuzumab                                                                                                                                                                                                                                                                           |
| <b>Peripheral edema</b>       | Dihydropyridine <b>Ca</b> <sup>2+</sup> channel blockers (eg, amlodipine)                                                                                                                                                                                                                                                                                                                       |
| <b>Torsades de pointes</b>    | Associated with agents that prolong QT interval: <b>M</b> ethadone, anti <b>A</b> rrhythmics (class IA, III), anti <b>B</b> iotics (eg, macrolides, fluoroquinolones), anti“ <b>C</b> ”ychotics (eg, ziprasidone), anti <b>D</b> epressants (eg, TCAs), anti <b>E</b> metics (eg, ondansetron), anti <b>F</b> ungals (eg, fluconazole) ( <b>M</b> emorize your <b>ABCDEF</b> )                  |

**Drug reactions—endocrine/reproductive**

| DRUG REACTION                | CAUSAL AGENTS                                                                                                                      | NOTES                                                                                           |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Adrenocortical insufficiency | HPA suppression secondary to chronic exogenous glucocorticoid use                                                                  | Abrupt withdrawal of exogenous glucocorticoids leads to adrenal crisis                          |
| Diabetes insipidus           | Lithium, demeclocycline                                                                                                            |                                                                                                 |
| Gynecomastia                 | Ketoconazole, cimetidine, spironolactone, GnRH analogs/antagonists, androgen receptor inhibitors, 5 $\alpha$ -reductase inhibitors |                                                                                                 |
| Hot flashes                  | SERMs (eg, tamoxifen, clomiphene, raloxifene)                                                                                      |                                                                                                 |
| Hyperglycemia                | Tacrolimus, protease inhibitors, niacin, HCTZ, glucocorticoids                                                                     | The people need High glucose                                                                    |
| Hyperprolactinemia           | Typical antipsychotics (eg, haloperidol), atypical antipsychotics (eg, risperidone), metoclopramide, methyl dopa, verapamil        | Presents with hypogonadism (eg, infertility, amenorrhea, erectile dysfunction) and galactorrhea |
| Hyperthyroidism              | Amiodarone, iodine, lithium                                                                                                        |                                                                                                 |
| Hypothyroidism               | Amiodarone, lithium                                                                                                                | I am lethargic                                                                                  |
| SIADH                        | Carbamazepine, Cyclophosphamide, SSRIs                                                                                             | Can't Concentrate Serum Sodium                                                                  |

**Drug reactions—gastrointestinal**

| DRUG REACTION                         | CAUSAL AGENTS                                                                                                                                                                                                                    | NOTES                                                                                                                                                                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute cholestatic hepatitis, jaundice | Macrolides (eg, erythromycin)                                                                                                                                                                                                    |                                                                                                                                                                      |
| Constipation                          | Antimuscarinics (eg, atropine), antipsychotics, opioids, non-dihydropyridine CCBs, ranolazine, amiodarone, aluminum hydroxide, loperamide, 5HT <sub>3</sub> receptor antagonist (ondansetron), vincristine                       |                                                                                                                                                                      |
| Diarrhea                              | Acamprosate, antidiabetic agents (acarbose, metformin, pramlintide), colchicine, cholinesterase inhibitors, lipid-lowering agents (eg, ezetimibe, orlistat), macrolides (eg, erythromycin), SSRIs, chemotherapy (eg, irinotecan) |                                                                                                                                                                      |
| Focal to massive hepatic necrosis     | <i>Amanita phalloides</i> (death cap mushroom), valproate, acetaminophen                                                                                                                                                         |                                                                                                                                                                      |
| Hepatitis                             | Rifampin, isoniazid, pyrazinamide, statins, fibrates                                                                                                                                                                             |                                                                                                                                                                      |
| Pancreatitis                          | Diuretics (eg, furosemide, HCTZ), glucocorticoids, alcohol, valproate, azathioprine                                                                                                                                              | Drugs generate a violent abdominal distress                                                                                                                          |
| Medication-induced esophagitis        | Potassium chloride, NSAIDs, bisphosphonates, ferrous sulfate, tetracyclines<br>Pills Not beneficial for food tube                                                                                                                | Usually occurs at anatomic sites of esophageal narrowing (eg, near level of aortic arch); caustic effect minimized with upright posture and adequate water ingestion |
| Pseudomembranous colitis              | Ampicillin, cephalosporins, clindamycin, fluoroquinolones, PPIs                                                                                                                                                                  | Antibiotics predispose to superinfection by resistant <i>C difficile</i>                                                                                             |

**Drug reactions—hematologic**

| DRUG REACTION                                         | CAUSAL AGENTS                                                                                              | NOTES                                                                                                                                                 |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agranulocytosis                                       | Dapsone, clozapine, carbamazepine, propylthiouracil, methimazole, ganciclovir, colchicine                  | Drugs can cause pretty major granulocytes collapse                                                                                                    |
| Aplastic anemia                                       | Carbamazepine, methimazole, NSAIDs, benzene, chloramphenicol, propylthiouracil                             | Can't make New blood cells properly                                                                                                                   |
| Direct Coombs ⊕ hemolytic anemia                      | Penicillin, cephalosporins, methyldopa                                                                     | Pooh classically munches on honey Coombs                                                                                                              |
| Drug Reaction with Eosinophilia and Systemic Symptoms | Phenytoin, carbamazepine, minocycline, sulfa drugs, allopurinol, vancomycin                                | T cell-mediated hypersensitivity reaction. Also known as drug-induced hypersensitivity syndrome (DIHS)<br>DRESSes partially cover my skin and viscera |
| Hemolysis in G6PD deficiency                          | Sulfonamides, dapsone, primaquine, aspirin, nitrofurantoin                                                 |                                                                                                                                                       |
| Megaloblastic anemia                                  | Hydroxyurea, Phenytoin, Methotrexate, Sulfa drugs                                                          | You're having a mega blast with PMS                                                                                                                   |
| Thrombocytopenia                                      | Heparin, quinidine, ganciclovir, vancomycin, linezolid                                                     |                                                                                                                                                       |
| Thrombotic complications                              | Combined oral contraceptives, hormone replacement therapy, SERMs<br>Testosterone supplements, epoetin alfa | Estrogen-mediated<br>↑ blood viscosity and platelet accumulation                                                                                      |

**Drug reactions—musculoskeletal/skin/connective tissue**

| DRUG REACTION                   | CAUSAL AGENTS                                                                                                           | NOTES                                                                |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Drug-induced lupus              | Hydralazine, procainamide, quinidine                                                                                    |                                                                      |
| Fat redistribution              | Protease inhibitors, glucocorticoids                                                                                    | Fat protects glutes                                                  |
| Gingival hyperplasia            | Cyclosporine, Ca <sup>2+</sup> channel blockers, phenytoin                                                              | Can Cause puffy gums                                                 |
| Hyperuricemia (gout)            | Pyrazinamide, thiazides, furosemide, niacin, cyclosporine                                                               | Painful tophi and feet need care                                     |
| Malignant hyperthermia          | Inhaled anesthetics (eg, isoflurane)                                                                                    | Individuals with ryanodine receptor mutation; antidote is dantrolene |
| Myopathy                        | Statins, fibrates, niacin, colchicine, daptomycin, hydroxychloroquine, interferon-α, penicillamine, glucocorticoids     |                                                                      |
| Osteoporosis                    | Glucocorticoids, depot medroxyprogesterone acetate, GnRH agonists, aromatase inhibitors, anticonvulsants, heparin, PPIs |                                                                      |
| Photosensitivity                | Sulfonamides, amiodarone, tetracyclines, fluoroquinolones                                                               | Sat for photo                                                        |
| Rash (Stevens-Johnson syndrome) | Anti-epileptic drugs (especially lamotrigine), allopurinol, sulfa drugs, penicillin                                     | Steven Johnson has epileptic allergy to sulfa drugs and penicillin   |
| Teeth discoloration             | Tetracyclines                                                                                                           | Teethracyclines                                                      |
| Tendon/cartilage damage         | Fluoroquinolones                                                                                                        |                                                                      |

**Drug reactions—neurologic**

| DRUG REACTION                        | CAUSAL AGENTS                                                                                                                                                                                                                                                                                                                           | NOTES                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Cinchonism                           | Quinidine, quinine                                                                                                                                                                                                                                                                                                                      | Can present with tinnitus, hearing/vision loss, psychosis, and cognitive impairment       |
| Parkinson-like syndrome              | Antipsychotics, metoclopramide                                                                                                                                                                                                                                                                                                          | Cogwheel rigidity of arm                                                                  |
| Peripheral neuropathy                | Platinum agents (eg, <b>cis</b> platin), <b>i</b> soniazid, vincristine, <b>pa</b> clitaxel, <b>p</b> henytoin                                                                                                                                                                                                                          | <b>Cis</b> , it's <b>very</b> <b>painful</b> <b>p</b> eripherally                         |
| Idiopathic intracranial hypertension | Corticosteroids, <b>da</b> nazol, vitamin <b>A</b> , growth hormones, <b>tetracyclines</b>                                                                                                                                                                                                                                              | <b>Crime</b> and <b>debt</b> <b>A</b> lways <b>g</b> row <b>h</b> ead <b>t</b> ension     |
| <b>Seizures</b>                      | <b>I</b> soniazid, <b>b</b> upropion, <b>i</b> mipenem/cilastatin, <b>tr</b> amadol                                                                                                                                                                                                                                                     | With <b>seizures</b> , <b>I</b> <b>bit</b> my tongue                                      |
| Tardive dyskinesia                   | Antipsychotics, metoclopramide                                                                                                                                                                                                                                                                                                          |                                                                                           |
| Visual disturbances                  | <b>T</b> opiramate (blurred vision/diplopia, haloes), <b>h</b> ydroxychloroquine (↓ visual acuity, visual field defects), <b>d</b> igoxin (yellow-tinged vision), <b>i</b> soniazid (optic neuritis), <b>i</b> vabradine (luminous phenomena), <b>P</b> DE-5 inhibitors (blue-tinged vision), <b>e</b> thambutol (color vision changes) | <b>T</b> hese <b>horrible</b> <b>drugs</b> <b>i</b> rritate <b>P</b> recious <b>e</b> yes |

**Drug reactions—renal/genitourinary**

| DRUG REACTION          | CAUSAL AGENTS                                                                                                                                      | NOTES                                 |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Fanconi syndrome       | Cisplatin, ifosfamide, expired tetracyclines, tenofovir                                                                                            |                                       |
| Hemorrhagic cystitis   | Cyclophosphamide, ifosfamide                                                                                                                       | Prevent by coadministering with mesna |
| Interstitial nephritis | Diuretics ( <b>P</b> ee), NSAIDs ( <b>P</b> ain-free), <b>P</b> enicillins and cephalosporins, <b>P</b> PIs, rifam <b>P</b> in, <b>sulfa</b> drugs | Remember the <b>5 P</b> 's            |
| Nephrotoxicity         | Cisplatin, aminoglycosides, amphotericin, vancomycin                                                                                               |                                       |

**Drug reactions—respiratory**

| DRUG REACTION      | CAUSAL AGENTS                                                                                                             | NOTES                                                                          |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Dry cough          | ACE inhibitors                                                                                                            |                                                                                |
| Pulmonary fibrosis | <b>M</b> ethotrexate, <b>n</b> itrofurantoin, <b>c</b> armustine, <b>bleo</b> mycin, <b>b</b> usulfan, <b>a</b> miodarone | <b>M</b> y <b>n</b> ose <b>c</b> annot <b>b</b> reathe <b>b</b> ad <b>a</b> ir |

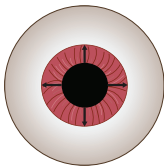
**Drug reactions—multiorgan**

| DRUG REACTION            | CAUSAL AGENTS                                                                                                                           | NOTES                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Antimuscarinic           | Atropine, TCAs, H <sub>1</sub> -blockers, antipsychotics                                                                                |                                                                     |
| Disulfiram-like reaction | 1st-generation <b>sulfonylureas</b> , <b>p</b> rocarbazine, certain <b>c</b> ephalosporins, <b>g</b> riseofulvin, <b>m</b> etronidazole | <b>S</b> orry <b>p</b> als, <b>c</b> an't <b>g</b> o <b>m</b> ingle |

Drugs affecting pupil size

↑ pupil size (mydriasis)

Anticholinergics (eg, atropine, TCAs, tropicamide, scopolamine, antihistamines)  
Indirect sympathomimetics (eg, amphetamines, cocaine, LSD), meperidine  
Direct sympathomimetics

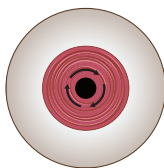


Radial muscle contraction  
( $\alpha_1$  receptor mediated)



↓ pupil size (miosis)

Sympatholytics (eg,  $\alpha_2$ -agonists)  
Opioids (except meperidine)  
Parasympathomimetics (eg, pilocarpine), organophosphates



Sphincter muscle contraction  
( $M_3$  receptor mediated)



Cytochrome P-450 interactions (selected)

Inducers (+)

**St. John's** wort  
**Phen**ytoin  
**Phen**obarbital  
**Mod**afinil  
**Nev**irapine  
**Rif**ampin  
**Griseo**fulvin  
**Carb**amazepine  
**Chronic alcohol** overuse

Substrates

**Theo**phylline  
**OCPs**  
**Anti**-epileptics  
**War**farin

Inhibitors (–)

**Sodium** valproate  
**Isonia**zid  
**Cime**tidine  
**Keto**conazole  
**Flu**conazole  
**Acute** alcohol overuse  
**Chloramphenicol**  
**Erythromycin/clarithromycin**  
**Sulfonamides**  
**Cipro**floxacin  
**Ome**prazole  
**Amio**darone  
**Rito**navir  
**Grapefruit juice**

**St. John's funny funny** (phen-phen) **mom never** refuses **greasy carbs** and **chronic alcohol**

**The OCPs** are **anti-war**

**SICK FACES** come when I **am** really drinking **grapefruit juice**

Sulfa drugs

Sulfonamide antibiotics, **Sulf**asalazine, **Probenecid**, **Furose**mide, **Acetazo**lamide, **Celecoxib**, **Thiazides**, **Sulfonylureas**.  
Patients with sulfa allergies may develop fever, urinary tract infection, Stevens-Johnson syndrome, hemolytic anemia, thrombocytopenia, agranulocytosis, acute interstitial nephritis, and urticaria (hives), and photosensitivity.

Scary Sulfa Pharm **FACTS**